

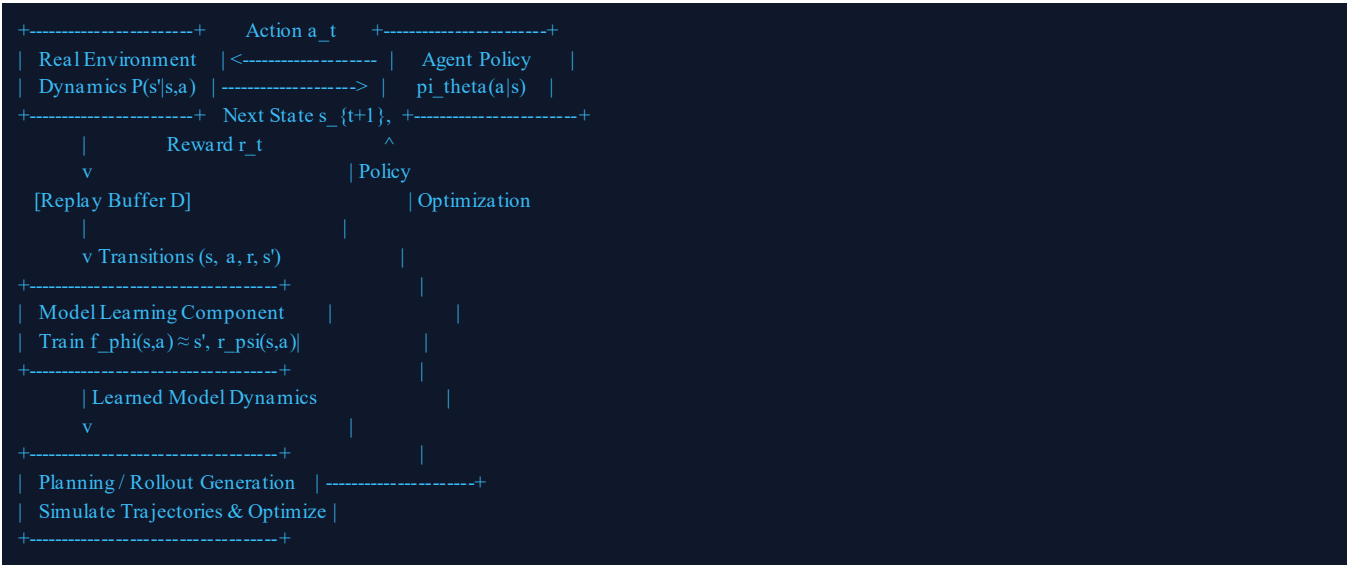
SIMATS ENGINEERING — ASSESSMENT TOOL 2

COURSE CODE:	MLA03	COURSE NAME:	Reinforcement Learning
COURSE OUTCOME:	CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)		
FACULTY:	Dr. G. A. SENTHIL	TOTAL MARKS:	50 Marks (40% Weightage) Duration: 4 Hrs

MODULE 1: MODEL-BASED RL FOUNDATIONS

Q1. Flowchart of Model-Based RL Workflow (Environment Interaction to Policy Optimization) [5 Marks]

Workflow Explanation: Model-Based Reinforcement Learning systematically alternates between real-world exploration, parameter estimation of the transition dynamics & reward functions, trajectory rollouts (simulated imagination), and policy/value improvement.



- **Step 1: Real-World Rollout:** Agent interacts with environment using current policy π_θ and stores transition tuples (s_t, a_t, r_t, s_{t+1}) into replay buffer D.
- **Step 2: Supervised Model Fitting:** Learn transition dynamics model f_ϕ by minimizing regression loss: $L(\phi) = E_{\{(s,a,s') \sim D\}} [\|f_\phi(s,a) - s'\|^2]$.
- **Step 3: Simulation & Optimization:** Generate virtual trajectories via f_ϕ and update policy parameters θ using either sampling-based trajectory planning or analytic policy gradients.

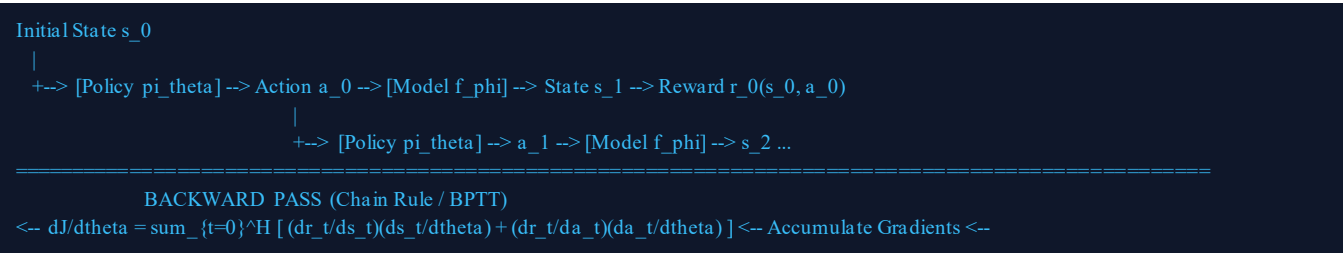
Q2. Concept Map: Model-Based RL vs. Model-Free RL Comparison [5 Marks]

Evaluation Dimension	Model-Free RL (e.g., PPO, SAC, DQN)	Model-Based RL (e.g., MBPO, PETS, MuZero)
Environment Knowledge	Direct trial-and-error; learns value/policy directly without explicit state transition dynamics.	Explicitly models state transitions $s_{t+1} \sim P_{\hat{}}(s' s,a)$ and rewards $R_{\hat{}}(s,a)$.
Sample Efficiency	Low (requires millions of real environment interactions to converge).	High (orders of magnitude fewer real samples; leverages virtual rollouts/imagination).
Computational Complexity	Lower compute per interaction step; fast policy updates.	High compute overhead due to iterative model fitting, trajectory forecasting, and planning.
Asymptotic Bias & Limits	Unbiased convergence to optimal policy; superior asymptotic performance.	Prone to model exploitation and compounding prediction error bias over long horizons.

MODULE 2: ANALYTIC GRADIENT COMPUTATION

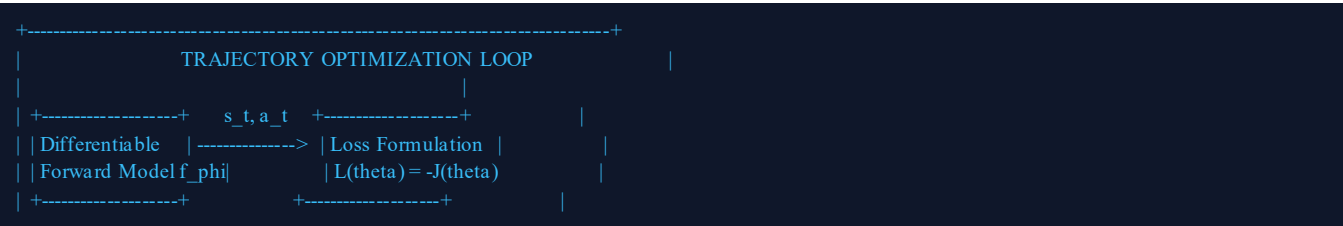
Q3. Flowchart: Analytic Gradient Computation & Backpropagation Through Dynamics [5 Marks]

Core Mechanism: When both the transition dynamics model $s_{t+1} = f_{\varphi}(s_t, a_t)$ and policy $a_t = \pi_{\theta}(s_t)$ are parameterized by differentiable neural networks, the gradient of the total expected return $\nabla_{\theta} J(\theta)$ can be computed analytically by unrolling the computation graph over horizon H and applying Backpropagation Through Time (BPTT).



- **Forward Trajectory Pass:** Starting from s_0 , roll out deterministic path: $a_t = \pi_{\theta}(s_t)$, $s_{t+1} = f_{\varphi}(s_t, a_t)$, $r_t = r(s_t, a_t)$ for $t = 0 \dots H-1$.
- **Recursive Chain Rule:** The state Jacobian propagates backward recursively: $ds_{t+1}/d\theta = (\partial f/\partial s_t)(ds_t/d\theta) + (\partial f/\partial a_t)[(\partial \pi/\partial s_t)(ds_t/d\theta) + \partial \pi/\partial \theta]$.
- **Parameter Update:** $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$ using first-order (Adam) or quasi-Newton gradient ascent.

Q4. Block Diagram: Policy Parameter Updates, Gradients, and Reward Optimization [5 Marks]



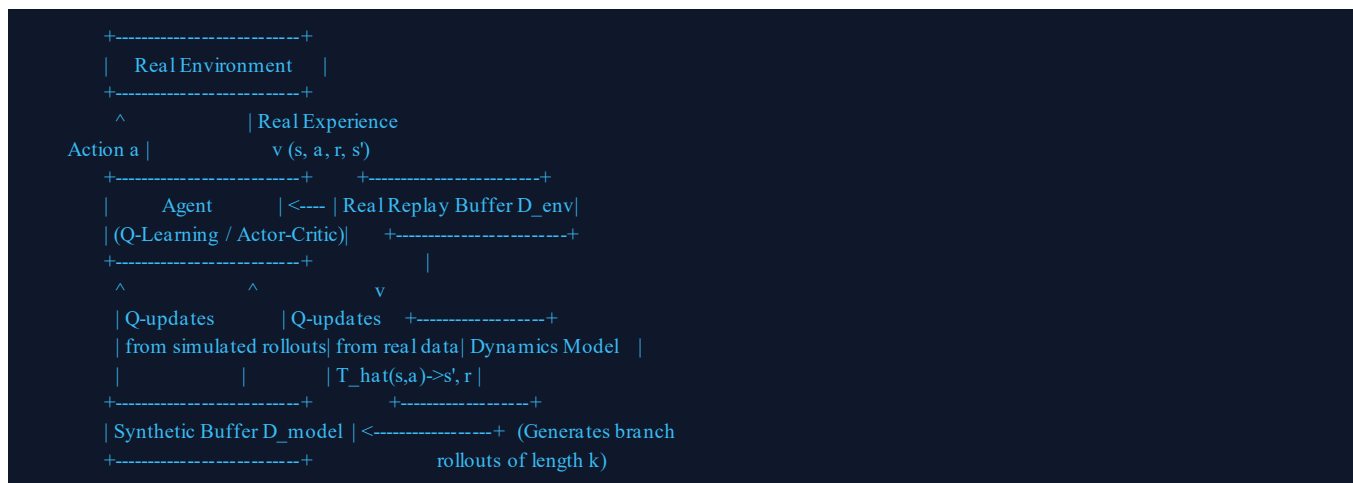
Q6. Concept Map: Exhaustive Search Planning vs. Sampling-Based Planning [5 Marks]

Dimension	Exhaustive Search (e.g., Minimax, BFS/DFS)	Sampling-Based Planning (e.g., MPC, CEM, MCTS)
Search Space Coverage	Evaluates all $ A ^H$ possible branches across the complete decision tree.	Samples a stochastic subset $K \ll A ^H$ of high-probability candidate paths.
Action Space Compatibility	Strictly limited to small discrete action spaces; intractable for continuous control.	Natively handles continuous, high-dimensional control spaces via Gaussian distributions.
Curse of Dimensionality	Suffers exponential computational explosion: $O(A ^H)$.	Scales polynomial/linearly with sample size: $O(K * H * T_{eval})$.
Optimality Guarantees	Guarantees exact globally optimal path across evaluated horizon H .	Asymptotically optimal or bounded sub-optimal depending on sample density K .

MODULE 4: MODEL-BASED DATA GENERATION

Q7. Block Diagram & Flowchart: Dyna-Architecture & Synthetic Data Generation [5 Marks]

Dyna Framework (Sutton): Unifies direct reinforcement learning (learning directly from real environment transitions) and indirect model-based learning (generating synthetic experience replay via the learned dynamics model).



- **Model Training:** Learns transition model T_{hat} on real dataset D_{env} .
- **Branched Rollouts:** Starts from real sampled states $s \sim D_{env}$, rolls out k-step simulated steps using policy π , and caches them in D_{model} .
- **Policy Update:** Samples mini-batches with mixing ratio $\beta D_{env} + (1-\beta) D_{model}$ to update Q-functions or Actor networks.

Q8. Concept Map: Advantages and Limitations of Model-Generated Training Data [5 Marks]

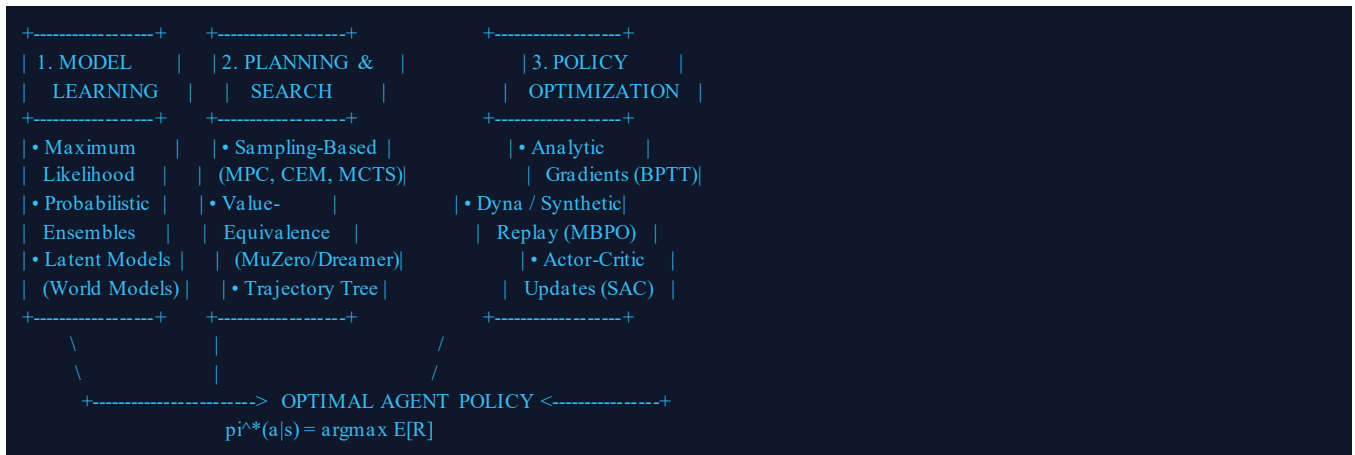
Category	Key Mechanisms	Impact on RL Performance
Advantages	• Sample Efficiency: Multiplies data volume 10x-100x. • Safe Exploration: Tests risky actions in simulation. • High Parallelism: GPU-batched vectorized rollouts.	Dramatically reduces costly and hazardous real-world hardware interactions.
Limitations & Pitfalls	• Compounding Model Errors: Errors accumulate exponentially over horizon H. • Model Exploitation: Policy exploits inaccurate high-reward regions. • Distribution Shift: Policy visits out-of-distribution states.	Can lead to severe policy degradation unless bounded by short rollouts (k -step) and probabilistic ensembles.

MODULE 5: VALUE-EQUIVALENCE & POLICY OPTIMIZATION (MBPO)

Q9. Block Diagram: Model-Based Policy Optimization (MBPO) Framework [5 Marks]

MBPO Framework (Janner et al.): Guarantees monotonic policy improvement by combining bootstrapped probabilistic neural network ensembles (to quantify epistemic uncertainty) with short k-step branched rollouts initialized from real environment states.

Q10. Comprehensive Concept Map: Unified Model-Based RL Framework [5 Marks]



Synthesis: This unified architecture illustrates how model learning, value-equivalent latent dynamics, and policy optimization integrate to overcome sample inefficiency and compounding error in complex decision-making tasks.

Code of Conduct:

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the Student: K. Balamurali

Date: 25-08-2026

RUBRICS FOR CALCULATION

Criteria	Weight (%)	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%	Highly accurate, in-depth, and insightful technical explanation.	Technically correct content with adequate depth and clarity.	Basic concepts presented with limited accuracy.	Major technical errors; poor understanding.
Problem Identification & Formulations	25%	Rigorously formulated algorithms with mathematical clarity and design flow.	Clearly defined approach with logical engineering flow.	Methodology or algorithmic steps are weak.	Unclear or incorrect statements.
Use of Tools & Visuals	20%	Advanced block diagrams, flowcharts, and structured concept maps.	Clear and effective diagrams used.	Limited or basic diagrammatic representation.	Minimal or uninformative visuals.
Communication & Organization	15%	Structured, professional, concise, and logically sequenced.	Organized and clear technical communication.	Understandable but lacks clean structure.	Poor organization; disjointed sections.
Professionalism & Integrity	10%	Complete adherence to academic integrity and rubric standards.	Professional presentation and adherence.	Minor formatting or integrity gaps.	Lacks professional formatting.